

Igor Lucindo Cardoso

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PROFESSIONAL SUMMARY

- PhD student specializing in Operations Research, leveraging Graph Theory and MIP for large-scale efficiency.
- Experienced in developing full-stack optimization tools using Python, CUDA, and Gurobi.

EDUCATION

Texas Tech University (TTU) Ph.D. in Industrial Engineering	2025 – Present GPA: 4.00/4.00
Getulio Vargas Foundation (FGV) M.A. Courses in Mathematical Modeling	2023 – 2024
Military Institute of Engineering (IME) B.S. in Electronic Engineering	2020 – 2024 GPA: 8.25/10.00

PUBLICATIONS

- **Cardoso, I.**, Teodoro, G., Validi, H. (2026). **A Polytime and Interpretable Approach for Solving Wordle**. Submitted to *Operations Research*.
- Gutierrez, R., **Cardoso, I.**, Validi, H. (2026). **Sport Scheduling to Minimize Travels at the FIFA World Cup 2026**. Submitted to *Optimization Letters*.
- **Cardoso, I.**, et al. (2024). **Comparing Pre-Trained Object Detection Models for Autonomous Grasp on Affordable Prosthetic Hands**. *IEEE MeMeA*.
- *Manuscripts in Progress*: **Cardoso, I.** and Validi, H. **On Solving the Network-Based Portfolio Optimization Problems**. *In prep*, 2026.

RESEARCH EXPERIENCE

Research Assistant, Texas Tech University (TTU) Developing novel algorithms for NP-hard combinatorial problems, leveraging GPU acceleration and decomposition techniques to solve large-scale instances. <i>Advisor: Dr. Hamidreza Validi</i> .	2025 – Present
Research Experience for Undergraduates, Memorial University (MUN) Developed autonomous grasping capabilities for a prosthetic hand by integrating machine learning, computer vision, and predictive modeling to drive system optimization. <i>Advisor: Dr. Vinicius P. da Fonseca</i> .	2023

HONORS & AWARDS

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| • TTU Tye Industrial Engineering Scholarship | 2026 | • 5th place, Petrobras Drone Challenge | 2021 |
| • INFORMS Student Chapter Award (Cum Laude) | 2025 | • Gold Medal, Brazilian Math Olympiad (OBMEP) | 2018 |
| • 1st place, Shell Eco-marathon "Road to 2050" | 2021 | | |

AFFILIATIONS & ACTIVITIES

- **INFORMS Chapter**: Seminars & research.
- **Robotics Club (RoboIME)**: Team member.
- **Journal Club**: Presenter in Ops Research.

SOFTWARE & PROJECTS

Wordle Optimization Solver Leveraged GPU-accelerated array ops for highly parallelized execution. [View App]	Python, CUDA, Decision Trees
FIFA World Cup 2026 Scheduler Interactive web app using MIP to optimize match scheduling and minimize travel. [View App]	Python, JS, MIP Solvers
Expense Splitter Optimization tool to minimize transactions required to settle shared expenses. [View App]	Python, Linear Programming
Computer Vision Rubik's Cube Solver Real-time web app mapping physical cubes to a 3D digital twin to generate CFOP solutions. [View App]	JS, OpenCV

SKILLS

Programming:	C/C++, Python, Matlab, Julia, Java, JavaScript, HTML, CSS, FPGA, SQL
Tools:	Gurobi, CUDA/GPU, Git, $\LaTeX$, Linux, NetworkX, ROS, OpenCV, Arduino, Firebase
Languages:	Portuguese (native), English (fluent), French (intermediate)